

RABBITMQ / .NET / IDEMPOTENCY

RabbitMQ redelivery in .NET Dedupe before you process.

A consumer that rejects a redelivered message instead of reprocessing it.

OrderConsumer.cs / dedupe guard

CSHARP

```
public async Task HandleAsync(BasicDeliverEventArgs ea, CancellationToken ct)
{
    var messageId = ea.BasicProperties.MessageId;
    var key = $"processed:{messageId}";

    var isNew = await cache.StringSetAsync(
        key, "1", TimeSpan.FromHours(24), When.NotExists);

    if (!isNew)
    {
        channel.BasicAck(ea.DeliveryTag, false);
        return;
    }

    await ProcessOrderAsync(ea.Body, ct);
    channel.BasicAck(ea.DeliveryTag, false);
}
```

NOTE

Brokers guarantee at-least-once delivery, not exactly-once. Track processed message IDs with a short TTL so a redelivery is a no-op, not a duplicate order.

N+1 hides until traffic finds it

Query once, not per row

Loading an order list should not fire one query per order for its line items.

BEFORE / lazy navigation per order

CSHARP 

```
var orders = await db.Orders
    .Where(o => o.CustomerId == id)
    .ToListAsync(ct);

foreach (var o in orders)
{
    // fires one query per order
    var total = o.Items.Sum(i => i.Price);
}
```

AFTER / project what you need

CSHARP 

```
var orders = await db.Orders
    .Where(o => o.CustomerId == id)
    .Select(o => new OrderSummary(
        o.Id, o.Items.Sum(i => i.Price)))
    .ToListAsync(ct);
```

NOTE

Lazy navigation looks correct in a unit test with three rows and falls over at production scale. Project or Include so EF Core issues one query, not N+1.

Same contract, two runtimes. Validate at the edge.

A CreateOrder DTO rejected before it reaches the domain layer, in Node and in .NET.

NestJS

TYPESCRIPT

```
class CreateOrderDto {  
  @IsUUID()  
  customerId: string;  
  
  @IsInt()  
  @Min(1)  
  quantity: number;  
}
```

.NET minimal API

CSHARP

```
public record CreateOrderDto(  
    [property: Required] Guid CustomerId,  
    [property: Range(1, int.MaxValue)] int Quantity  
);
```

NGINX / RATE LIMITING

A limit that trips itself. `limit_req` needs burst.

Without burst and nodelay, a page load with six assets rate-limits itself.

No burst

NGINX 

```
limit_req_zone $binary_remote_addr
    zone=api:10m rate=5r/s;

location /api/ {
    limit_req zone=api;
}
```

-  Six parallel requests trip the limit for one user
-  503s show up in real traffic, not just load tests

Zone + burst

NGINX 

```
limit_req_zone $binary_remote_addr
    zone=api:10m rate=5r/s;

location /api/ {
    limit_req zone=api burst=12 nodelay;
}
```

-  Absorbs a page's worth of concurrent calls
-  Still caps sustained abuse at 5r/s

NOTE

A tight rate without burst rejects the same client's own concurrent asset requests, not just abusive traffic. Zone size and burst are what make the limit usable.

DOCKER / COMPOSE / HEALTHCHECKS

Healthy doesn't mean started

`depends_on: service_healthy`

Four services, four healthchecks — the API waits for real readiness, not just a running container.

postgres

YAML

```
postgres:
  image: postgres:16
  healthcheck:
    test: ["CMD-SHELL", "pg_isready -U app"]
    interval: 5s
    retries: 5
```

redis

YAML

```
redis:
  image: redis:7
  healthcheck:
    test: ["CMD", "redis-cli", "ping"]
    interval: 5s
    retries: 5
```

rabbitmq

YAML

```
rabbitmq:
  image: rabbitmq:3-management
  healthcheck:
    test: ["CMD", "rabbitmq-diagnostics", "ping"]
    interval: 10s
    retries: 5
```

api

YAML

```
api:
  build: .
  depends_on:
    postgres:
      condition: service_healthy
    redis:
      condition: service_healthy
```

NOTE

`condition: service_started` only waits for the container to launch.
`service_healthy` waits for the healthcheck to pass, which is what dependent services actually need.

ANGULAR / DI / SIGNALS

Same DI, less ceremony. `inject()` in Angular

Constructor injection still works; the `inject()` function removes the boilerplate around it.

Constructor injection

TYPESCRIPT



```
export class OrderListComponent {  
  constructor(  
    private orders: OrderService,  
    private auth: AuthService,  
  ) {}  
}
```

- ✗ Verbose for components with several dependencies
- ✗ Parameter order matters for readability

`inject()`

TYPESCRIPT



```
export class OrderListComponent {  
  private orders = inject(OrderService);  
  private auth = inject(AuthService);  
}
```

- ✓ Reads top to bottom as plain field assignment
- ✓ Works in functional guards and resolvers too

NOTE

`inject()` must run in an injection context — a constructor, a field initializer, or a factory function — not inside a later callback or `setTimeout`.

Name the role, not the colour.

Semantic design tokens

A stylesheet that says "accent" picks up the next palette release; one that says #2a5aa8 never will.

Raw values

CSS 

```
.button {  
  background: #2a5aa8;  
  color: #ffffff;  
  border: 1px solid #87816f;  
}
```

Semantic tokens

CSS 

```
.button {  
  background: var(--mk-color-accent);  
  color: var(--mk-color-on-accent);  
  border: 1px solid var(--mk-color-border-control);  
}
```

NOTE

Primitives hold raw values and never reach CSS; semantic tokens hold meaning and are all a site sees. Renaming one breaks consumers, changing its value does not.

CSS / DARK MODE

Follow the OS, allow an override Light and dark from one file

prefers-color-scheme picks the mode by default; a data-theme attribute on <html> forces one.

tokens.css

CSS

```
:root {
  --mk-color-bg: #F2F0EB;
  color-scheme: light;
}

@media (prefers-color-scheme: dark) {
  :root:not([data-theme="light"]) {
    --mk-color-bg: #0F141B;
    color-scheme: dark;
  }
}

:root[data-theme="dark"] {
  --mk-color-bg: #0F141B;
  color-scheme: dark;
}
```

- No attribute: the OS preference decides
- color-scheme switches native controls too

index.html

HTML

```
<html data-theme="dark">
  <head>
    <link
      rel="stylesheet"
      href="/v1/index.css"
    />
  </head>
  <body>
    ...
  </body>
</html>
```

- data-theme="light" or "dark" forces a mode
- Remove it to follow the OS again